

YITIAN WANG



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SUMMARY

Postdoctoral Fellow at The University of Texas at Austin with a Ph.D. in Electrical Engineering and expertise in thin-film deposition (PVD/CVD), crystal growth, materials characterization, and process optimization. Experienced in DOE-based process development, hardware troubleshooting, and large-scale data analysis using Python and MATLAB. Led multi-institutional experiments at Oak Ridge National Laboratory and published 6 first-author and 3 co-authored peer-reviewed journal articles. Strong foundation in semiconductor device physics and CMOS process flows through graduate-level coursework.

EDUCATION

University of California, Riverside, CA Ph.D. in Electrical and Computer Engineering Awards: MRS 2023 Highly Commended Student Talk Award; Dissertation Completion Fellowship Coursework: Semiconductor Device Physics, CMOS Fabrication, Thin-Film Technology	<i>Mar. 2021 - Jun. 2026</i>
Columbia University in the City of New York, NY Master of Science, Material Science and Engineering	<i>Aug. 2019 - Feb. 2021</i>
Beijing Normal University, CN Bachelor of Science, Physics Awards: 'Jingshi' Scholarship (2016 - 2018)	<i>Sep. 2015 - Jun. 2019</i>

SKILLS

Process & Deposition	PVD / Magnetron Sputtering, CVD, Floating-Zone Crystal Growth, DOE
Characterization	SEM, TEM, XRD, AFM, Raman, FTIR, Inelastic Neutron Scattering, PPMS, DSC, TGA, Four-Probe Transport
Software	Python, MATLAB, PyTorch, R, SQL, GSAS-II, Mantid, ImageJ, SolidWorks, Quantum ESPRESSO, VASP, COMSOL
Electrochemistry	Prototype Cell Assembly, Separator Fabrication, EIS, Galvanostatic Cycling

RESEARCH AND ENGINEERING EXPERIENCE

Postdoctoral Fellow <i>The University of Texas at Austin</i>	<i>Jul. 2026 - Present</i> <i>Austin, TX</i>
Conducting postdoctoral research in materials and energy-related engineering.	
Graduate Research Assistant — Process Development & Characterization <i>University of California, Riverside</i>	<i>Jun. 2021 - Jun. 2025</i> <i>Riverside, CA</i>
- Designed and executed DOE-based experiments to optimize crystal growth and thin-film deposition processes for solid-state electrolyte materials. - Performed systematic hardware characterization on floating-zone furnaces and deposition systems, troubleshooting process issues and implementing parameter improvements. - Measured and analyzed film and bulk material properties (XRD, SEM, Raman, FTIR, four-probe transport) to establish process-property correlations. - Built automated data analysis and fitting frameworks in MATLAB and Python, generating technical reports and presentations for cross-functional teams. - Managed vendor relationships (Thermo Fisher, Sigma-Aldrich, MTI) and mentored junior graduate students on experimental techniques.	
Experiment Lead — Neutron Scattering Characterization <i>Oak Ridge National Laboratory</i>	<i>Jan. 2022 - Jul. 2023</i> <i>Oak Ridge, TN</i>

- Led complex characterization experiments on solid electrolytes using TAX (HFIR) and ARCS (SNS) instruments, coordinating with beamline scientists to optimize experimental parameters.
- Developed automated data processing and analysis workflows in Mantid and Python, reducing analysis turnaround and improving reproducibility.
- Generated technical documentation and coordinated across 5+ institutions including ORNL, ETH Zürich, and TU Wien.

Graduate Researcher — Battery Materials & Process Engineering

Columbia University

Sep. 2019 - Jan. 2021

New York, NY

- Developed composite separator fabrication processes using solution processing of γ -C₃N₄/PVDF, optimizing film uniformity and mechanical properties.
- Assembled prototype coin cells and performed electrochemical characterization (EIS, galvanostatic cycling) to evaluate process quality and device performance.
- Conducted first-principles simulations (Quantum ESPRESSO) to predict defect behavior and guide process design for olivine-type electrode materials.

Student Researcher — Hardware Design & Characterization

Beijing Normal University

Sep. 2018 - May 2019

Beijing, CN

- Designed precision uniaxial-strain devices in SolidWorks for thin-film sample characterization.
- Performed systematic transport measurements using PPMS, developing testing protocols for temperature-dependent material studies.

PROFESSIONAL MEMBERSHIPS

- Sigma Xi, The Scientific Research Honor Society — Elected Full Member (2026)
- American Physical Society (APS) — Member (2025 – 2026)
- Materials Research Society (MRS) — Student Member (through 2027)
- The Society for Collegiate Leadership & Achievement (SCLA) — Member (2026)

SELECTED PUBLICATIONS

1. **Y. Wang**, H. Wang, X. Chen. “Thermal transport mechanisms in electrolytes for all-solid-state batteries.” *Appl. Phys. Rev.*, 2026. [\[Link\]](#)
2. **Y. Wang** et al. “Origin of intrinsically low thermal conductivity in a garnet-type solid electrolyte.” *PRX Energy*, 2025. [\[Link\]](#)
3. **Y. Wang** et al. “Glass-like thermal transport in perovskite Li-ion conductor.” *Chem. Commun.*, 2025 (Front Cover). [\[Link\]](#)
4. **Y. Wang** et al. “Low thermal conductivity of NaSICON-type solid electrolyte.” *Tungsten*, 2025. [\[Link\]](#)
5. **Y. Wang** et al. “Thermal properties of Li-ion conducting garnet solid electrolyte.” *J. Mater. Chem. A*, 2024. [\[Link\]](#)
6. S. Li, S. Guo, **Y. Wang**, et al. “Enhanced magnon thermal transport in Y-doped spin-ladder Sr_{14-x}Y_xCu₂₄O₄₁.” *J. Appl. Phys.*, 2024. [\[Link\]](#)
7. S. Guo, H. Li, X. Bai, **Y. Wang**, et al. “Size-dependent magnon thermal transport in a nanostructured quantum magnet.” *Cell Rep. Phys. Sci.*, 2024. [\[Link\]](#)
8. Y. Xu, Z. Barani, P. Xiao, S. Sudhindra, **Y. Wang**, et al. “Crystal structure and thermoelectric properties of layered van der Waals semimetal ZrTiSe₄.” *Chem. Mater.*, 2022. [\[Link\]](#)